

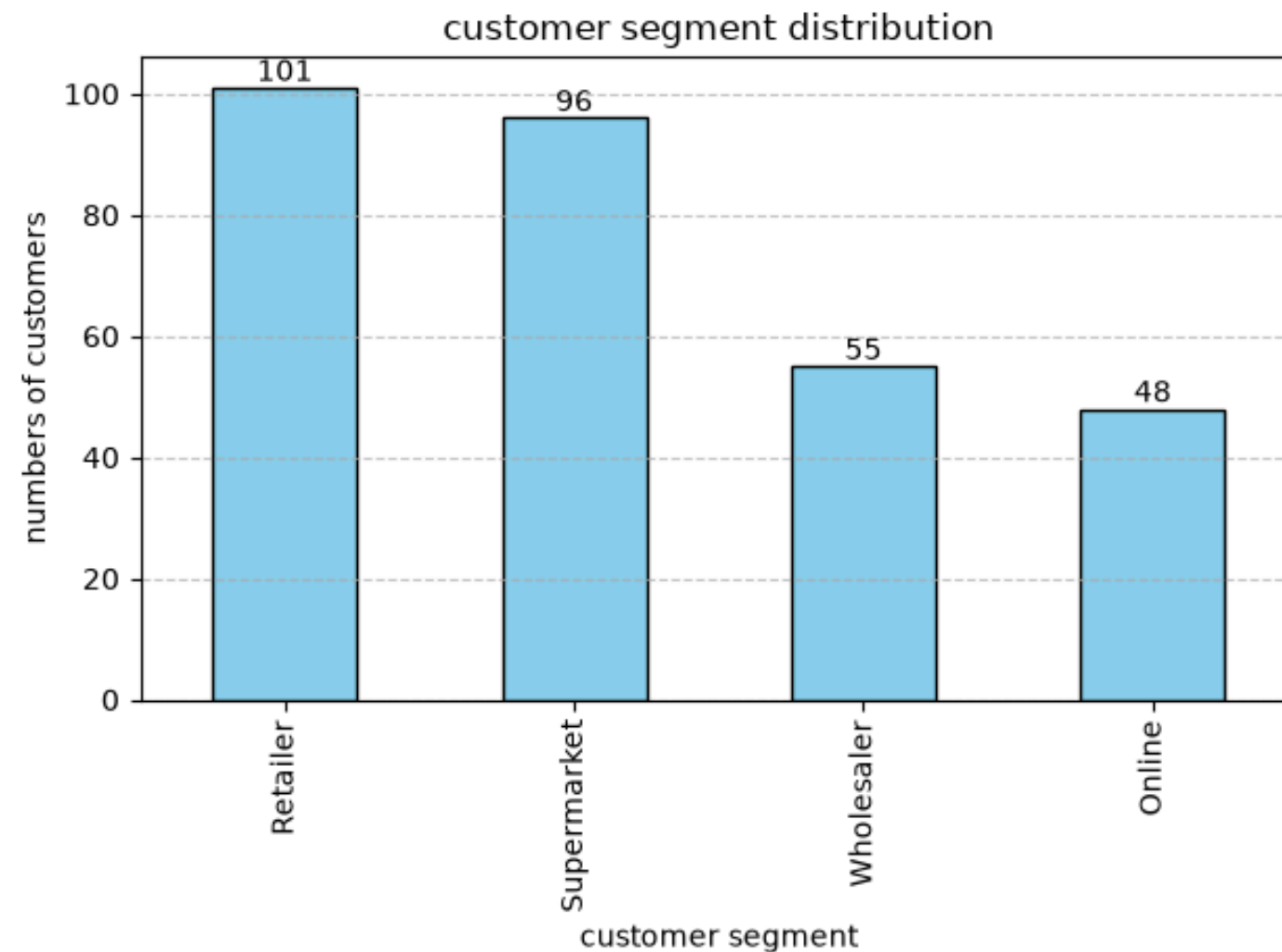


FMCG Sales & Supply Chain Analytics

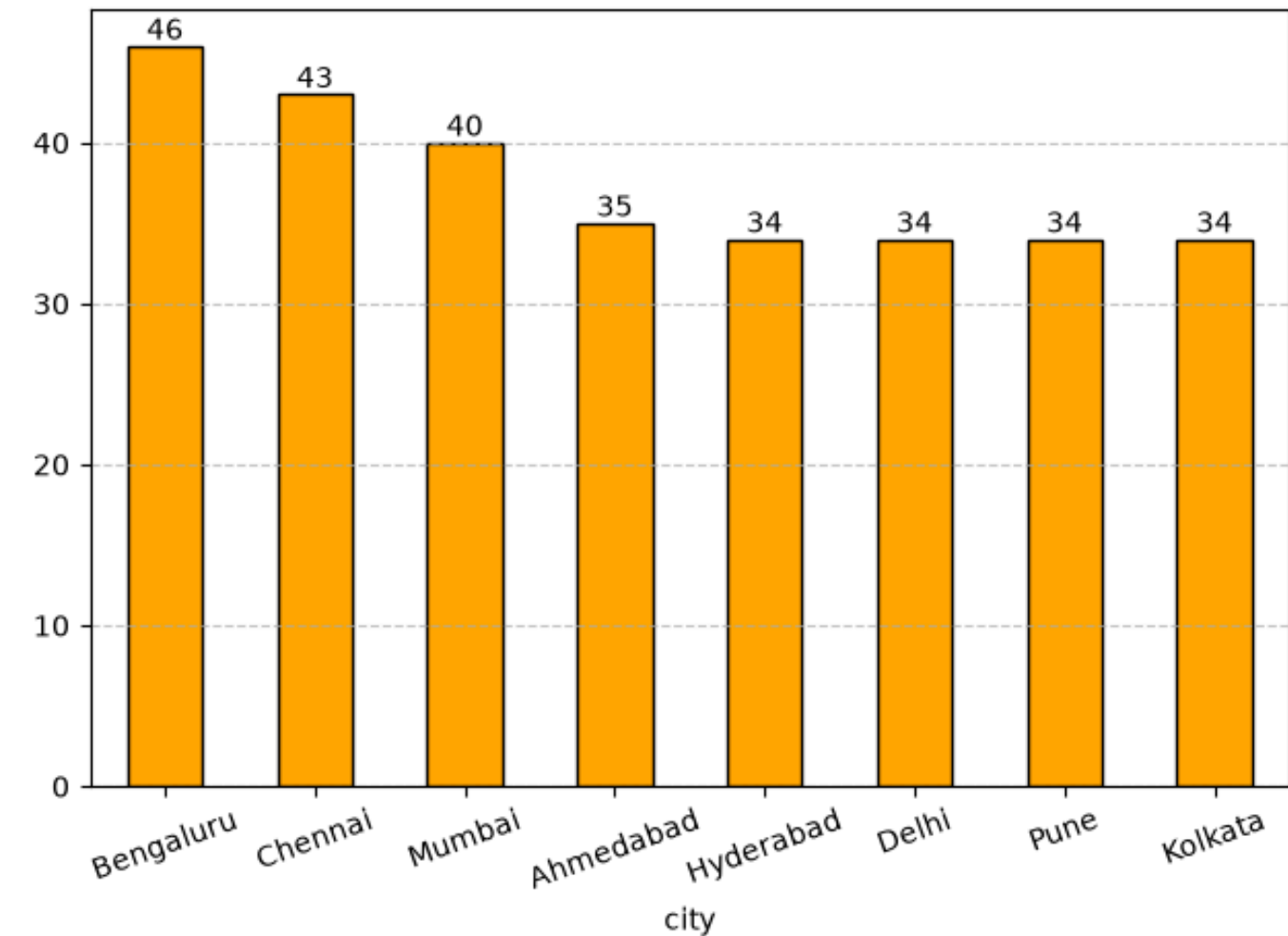
- **Business Insights & Recommendations**
- **Python-Based Exploratory Data Analysis**
- **Tools : Python | Pandas | NumPy | Matplotlib | Seaborn**

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Customers Distribution across segment and cities :

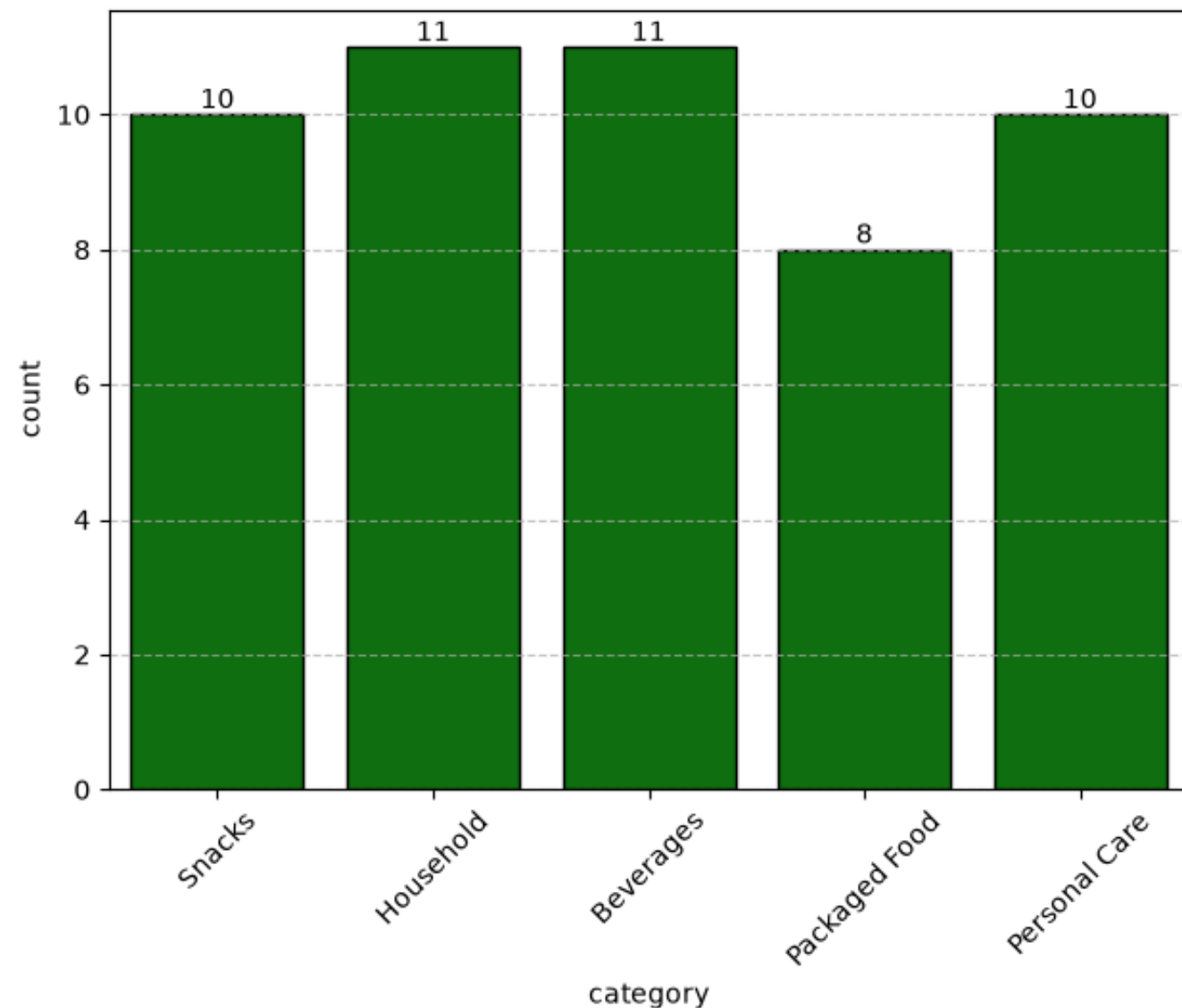


- **Retailers and Supermarkets** represent the largest customer segments, indicating strong dependence on offline channels.
- **Wholesalers contribute moderately**, while Online customers form the smallest segment, highlighting an opportunity to expand digital sales.



- **Bengaluru** has the highest customer presence, followed by **Chennai and Mumbai**.
- Customer distribution is relatively balanced across the remaining major cities.
- **Business Opportunity:** Strengthen online customer acquisition while maintaining strong offline channels.

Product Category And Selling Price Distribution :

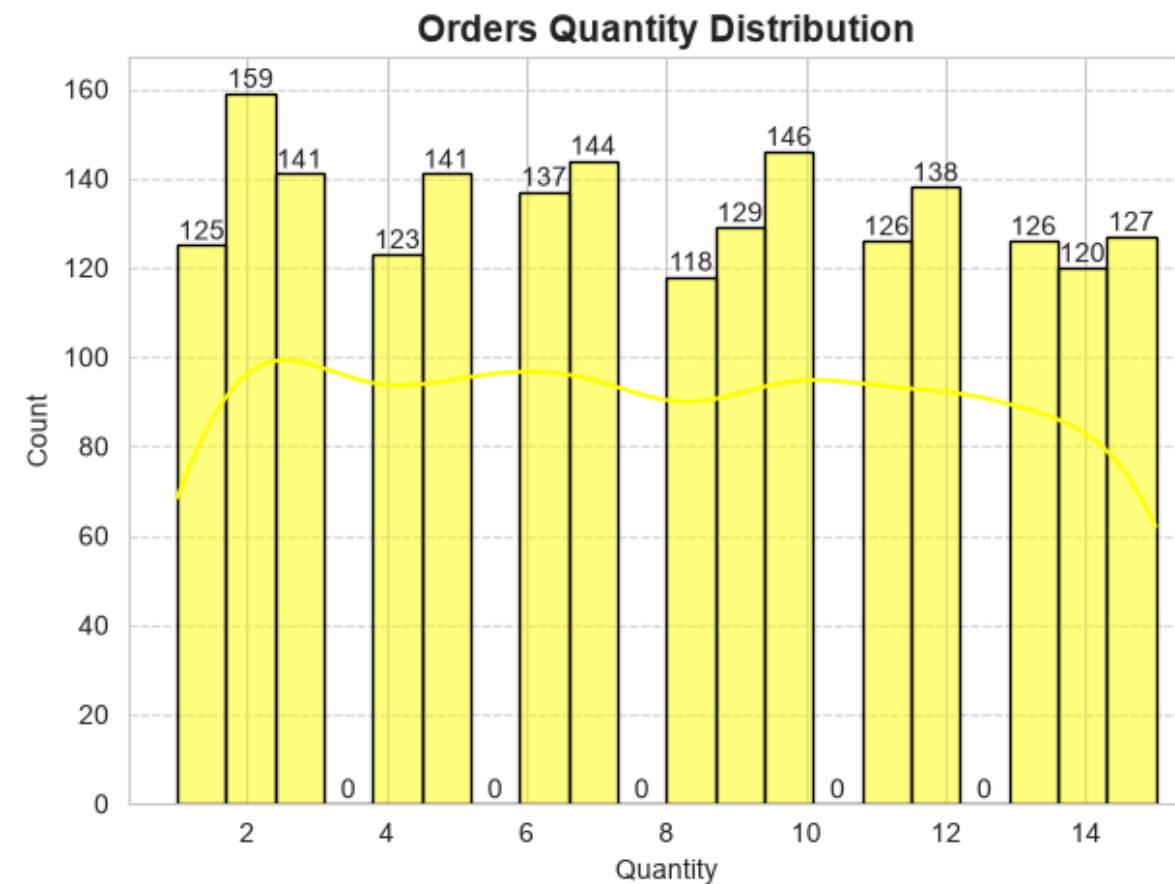
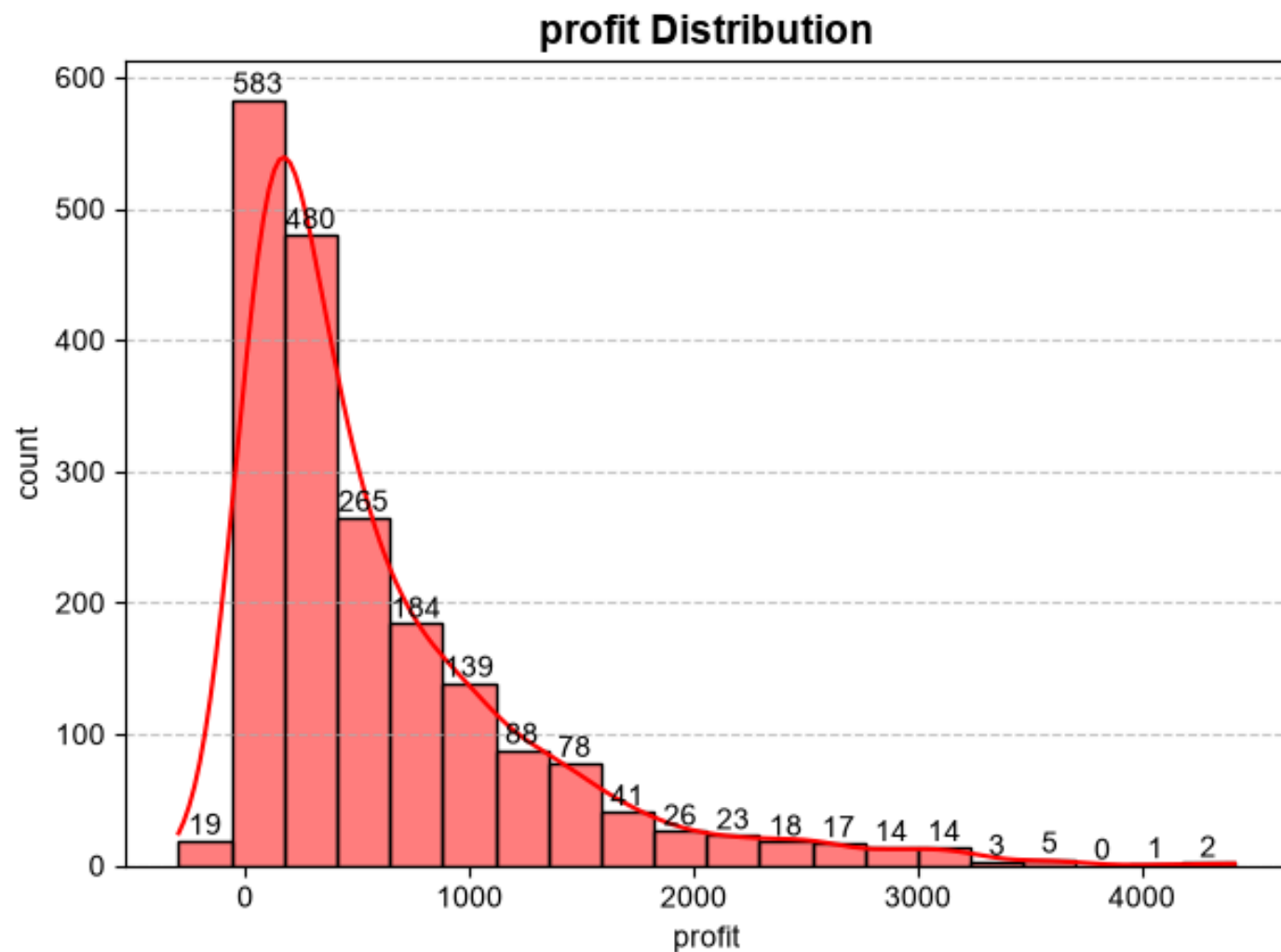


- **Household and Beverages** have the largest product assortment, with **11 products each**.
- Packaged Food has the smallest assortment, with 8 products.



- Most products are priced between **₹100–₹300**, indicating a portfolio concentrated in affordable price ranges.
- Very few products are priced above **₹600**.
- The product portfolio is primarily positioned toward cost-sensitive customers.

Profitability & Order Distribution :



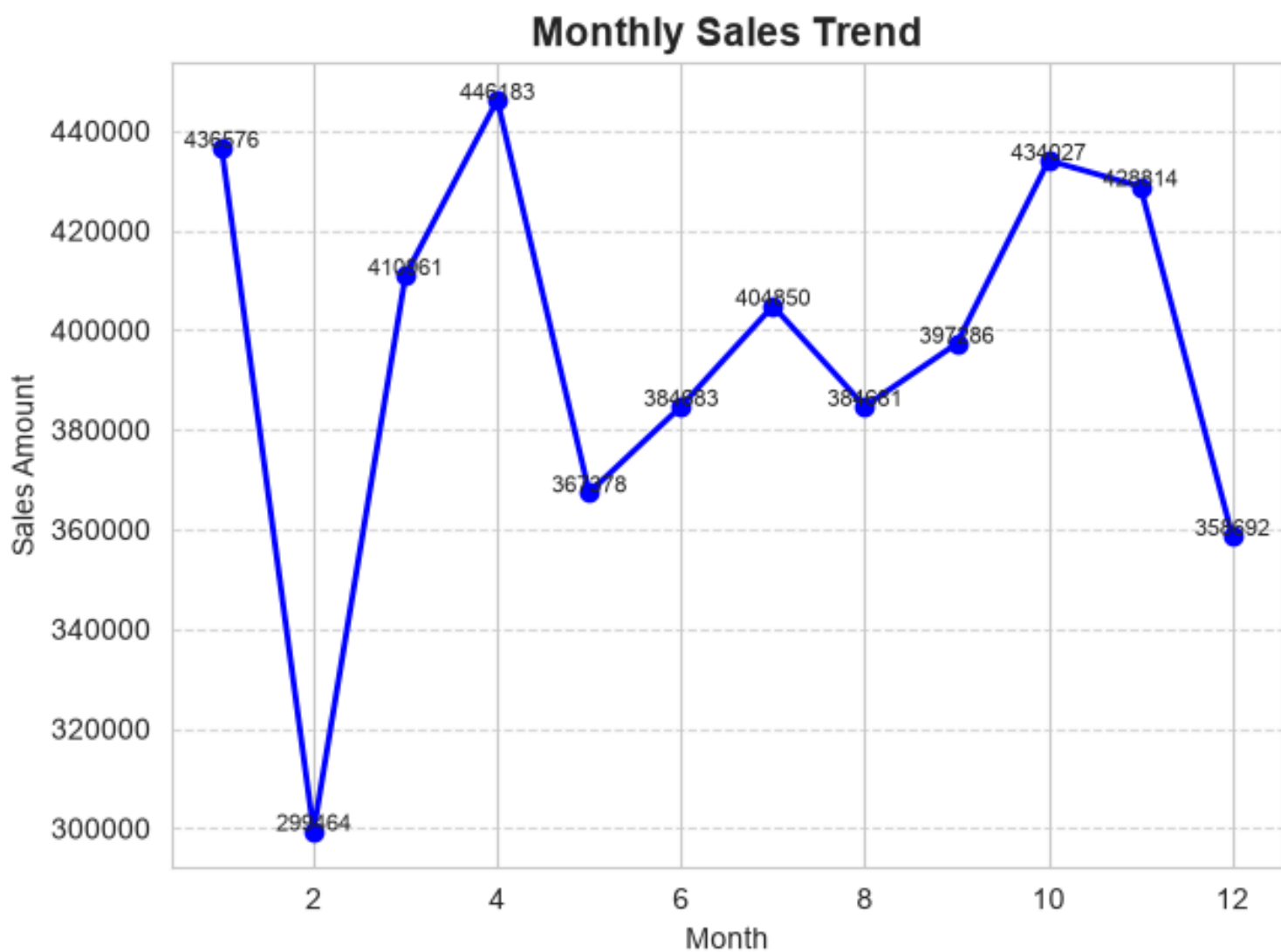
Category	Profit
Beverages	₹2.81 lakh
Households	₹2.19 lakh
Packaged food	₹2.09 lakh
Personal care	₹2.03 lakh
Snacks	₹ 2.85 lakh

- Out of **2000 orders**, Only **2.54%** of orders resulted in losses, indicating that the overall order portfolio is largely profitable.
- Loss-making orders should be further analyzed by product and category to identify their drivers.

- **Average order quantity** is approximately 8 units, with orders distributed between 1 and 15 units.
- This provides a useful baseline for inventory and packaging planning.

- **Snacks** generate the highest total profit, closely followed by **Beverages**.
- **Personal Care** generates the lowest profit among the **categories**.

Monthly Sales and Category Insights :



- **April** recorded the highest monthly sales, followed by **October and July**.
- February recorded the lowest sales, indicating potential seasonal variation in demand.

Top 10 products by revenue:

- **P009 and P002** are the leading revenue-generating products and contribute significantly to overall sales.
- Lower-performing products such as **P003 and P007** may require targeted pricing or promotional strategies.
- Revenue is concentrated around a small group of high-performing products, making inventory availability for these products critical.

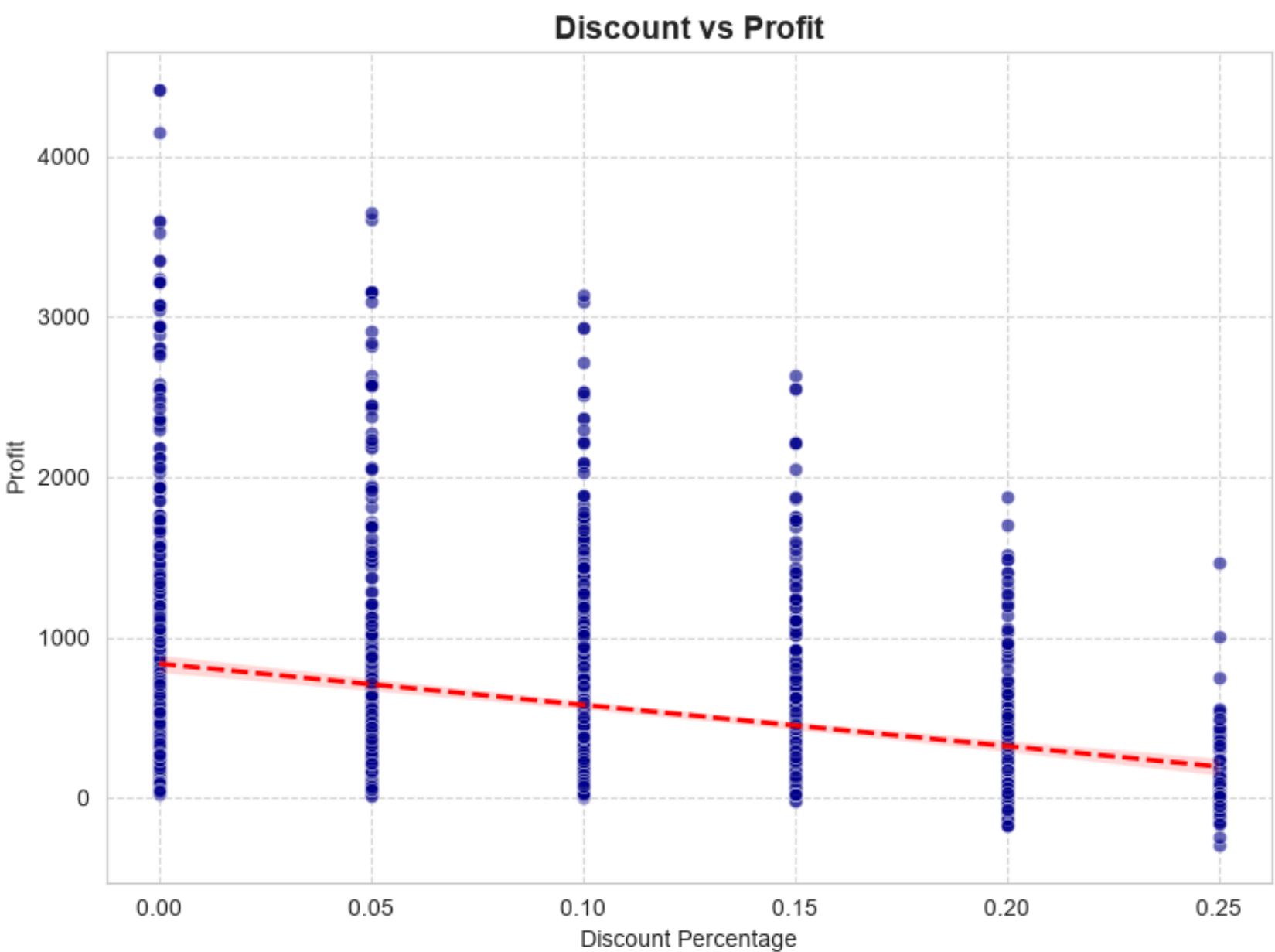
Category	Revenue
Beverages	₹10.90 lakh
Households	₹8 lakh
Packaged food	₹7 lakh
Personal care	₹ 9 lakh
Snacks	₹10.43 lakh

- **Beverages** generate the highest category revenue, followed by **Snacks**.
- **Packaged Food** generates the lowest category revenue among the five **categories**.

Category Performance & Discount Effect on Profit :

Category	Revenue	Profit	Profit Margin
Beverages	₹10.90 lakh	₹ 2.81 lakh	25.87%
Households	₹8 lakh	₹ 2.19 lakh	24.84%
Packaged food	₹7 lakh	₹2.09 lakh	27.18%
Personal care	₹ 9 lakh	₹ 2.03 lakh	21.05%
Snacks	₹10.43 lakh	₹ 2.85 lakh	27.36%

- **Snacks and Packaged Food** have the highest profit margins at approximately **27%**.
- **Personal Care** has the lowest margin at approximately 21%, despite generating relatively **high revenue**.
- This indicates a potential opportunity to review pricing and cost structure in **Personal Care**.



- **There is a weak negative correlation (-0.28)** between discount percentage and profit, suggesting that higher discounts tend to reduce profitability. Regression line → shows overall negative trend (**discount ↑ → profit ↓**).

Revenue, Profit & Loss Breakdown :

Discount_pct	Profit_mean
0.00	810.86
0.05	748.09
0.10	593.67
0.15	445.88
0.20	304.43
0.25	174.30

- **Higher discounts are moderately associated with lower profit.** The company should further evaluate discount levels using profitability and sales-volume analysis before setting discount limits.

Customer_segment	revenue	Profit
Online	₹ 7 lakh	₹1.83 lakh
Retailer	₹15.44 lakh	₹ 3.86 lakh
Supermarket	₹15.84 lakh	₹3.96 lakh
Wholesaler	₹ 9.24 lakh	₹2.33 lakh

- **Supermarket** is the largest customer segment by both revenue and absolute profit, generating approximately ₹**15.84** lakh in revenue and ₹3.97 lakh in profit.
- Online generates the lowest revenue but has the highest profit margin among the customer segments, indicating potential for profitable growth.

Category	Loss
Households	₹-648.98
Packaged food	₹-752.34
Personal care	₹-1386.27
Snacks	₹-286.04

- **Personal Care** records the highest category-level loss at approximately ₹**1,386**.
- This category should be prioritized for further investigation into pricing, discounts and cost structure.

Recommendations :

- **Grow Online** : Explore strategies to expand the Online segment while maintaining its relatively high profitability.
- **Protect High-Revenue Products** : Prioritize inventory availability and promotions for leading products such as P009 and P002.
- **Optimize Personal Care** : Review pricing, discounts and cost structure to improve its relatively low profit margin and category-level losses.
- **Improve Discount Management** : Evaluate discount levels using both profit and sales volume before establishing discount thresholds.
- **Optimize Inventory** : Prioritize fast-moving products and monitor slow-moving products to reduce potential overstock risk.